

Taking Dual Shortcut

Primal (Max)	Dual (min)
i th con $\leq$	i th var $\geq$
i th con $\geq$	i th var $\leq$
i th con $=$	i th var unrestricted
j th var $\geq$	j th con $\geq$
j th var $\leq$	j th con $\leq$
j th var unrestricted	j th con $=$

Note that the costs of the primal are now the b of the dual and that the b of the primal are now the costs of the dual.

When trying to figure out how taking the dual of parametric LP's think about when a variable appears in more than one constraint, in which case those constraints are all important for that variable in the dual. If a variable x appears in constraints $a_1, \dots, a_l$ then those constraint variables $a_1, \dots, a_l$ appear in the dual in the constraint corresponding to the variable x . Additionally, these constraint variables appear in this specific constraint with the same scalar multiple in which x appeared in the individual constraints.

Lagrangian Relaxtion in QUBO

$$L(\lambda) = \min c^T x - \lambda(f^T x - g)$$

Boolean Condition	Linear Constraint	Quadratic Constraints
$x \vee y$	$x + y \geq 1$	$1 - x - y + xy = 0$
$x \rightarrow y$	$x \leq y$	$x - xy = 0$
$\neg(x \wedge y)$	$x + y \leq 1$	$xy = 0$
$x \wedge y$	$x + y = 2$	$1 - xy = 0$
$x \oplus y$	$x + y = 1$	$(1 - x - y)^2 = 1 - x - y + 2xy$
$\oplus(x)$	$\sum x = 1$	$(1 - \sum x)^2 \equiv 1 - \sum_i x_i + 2 \sum_{0 \leq i < j \leq n} x_i x_j$

Finding Basic Feasible Solutions

A bfs $\bar{x}$ is found by finding a basis B of A and inverting B then multiplying by b to get the vector x where any variables not in the basis b are simply set to 0.

$$\pi \ \& \ c^\pi$$

For a basis B of A :

$$\pi = (B^{-1})^T C_B$$

$$C^\pi = C - A^T \pi$$

Proving Optimality

Either show that $c^\pi \geq 0$ or that complementary slackness is 0 ie $y(Ax - b) = x(A^T y - c) = 0$. Note complementary slackness only applies to non equality constraints as the complementary slackness would be 0 for any feasible solution.

Taking the Dual of HW 2 Q5

$$\min \sum y_h$$

$$\text{s.t. } x_{Ih} \leq y_h, (I \in A, h = 1, \dots, n)$$

$$x_{Ih} + x_{Jh} \leq 1, ((I, J) \in c(A), h = 1, \dots, n)$$

$$\sum_h x_{Ih} = 1, (I \in A)$$

$$x_{Ih}, y_h \leq 1, (I \in A, h = 1, \dots, n)$$

so

$$\min \sum y_h$$

$$\text{s.t. } x_{Ih} - y_h \leq 0, (I \in A, h = 1, \dots, n)$$

$$x_{Ih} + x_{Jh} \leq 1, ((I, J) \in c(A), h = 1, \dots, n)$$

$$\sum_h x_{Ih} = 1, (I \in A)$$

$$x_{Ih}, y_h \leq 1, (I \in A, h = 1, \dots, n)$$

$$x_{Ih}, y_h \geq 0, (I \in A, h = 1, \dots, n)$$

Dual

Lets denote the first constraints as our districts $d_{ij}, i \in A, j \in n$ and the next set as conflicts $\beta_{ikj}, (i, k) \in c(A), k \in n$, the next as appearances $\alpha_i, i \in A$ and lastly upper bounds as $ux_{Ih}, I \in A, h \in n$ and $uy_h, h \in n$

Thus our dual is:

$$\begin{aligned} \max \quad & \sum \beta + \sum \alpha + \sum ux_{Ih} + \sum uy_h \\ \text{s.t.} \quad & d_{Ih} + \sum_{(I,J) \in c(A)} \beta_{IJh} + \alpha_I + ux_{Ih} \leq 0, (I \in A, h = 1, \dots, n) \\ & - \sum_{i \in A} d_{ij} + uy_j \leq 1, (j = 1, \dots, n) \\ & d \leq 0, \beta \leq 0, u \leq 0 \end{aligned}$$

MST as ILP

Given an undirected graph $G(V, E)$ with edge weights w we can create an LP as follows where $m = |E|, n = |V|$:

$$\begin{aligned} \min \quad & \sum_{(i,j) \in E} w_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j: (i,j) \in E} x_{ij} \geq 1, (i = 1, \dots, n) \\ & \sum_{(i,j) \in E} x_{ij} = n - 1 \\ & x_{ij} \in \{0, 1\}, ((i, j) \in E) \end{aligned}$$

Vertex Cover as ILP

Given an undirected graph $G(V, E)$ with edge weights w we can create an LP as follows where $m = |E|, n = |V|$:

$$\begin{aligned} \min \quad & \sum_{i \in V} x_i \\ \text{s.t.} \quad & x_i + x_j \geq 1, ((i, j) \in E) \\ & x_i \in \{0, 1\}, \forall i \in V \end{aligned}$$

SDP LP formulation

Suppose we have original LP as: $\min cx$

s.t. $\bar{A}x = b_m$

$$x \geq 0$$

with $|x| = n$

So the SDP is

$$b = \begin{pmatrix} b_m \\ b_z \end{pmatrix}, X = xI, C = cI, A = \{rI\} \cup \{Z\}$$

And $b_z = n^2 - n$ list of 0's and Z is a list of $n^2 - n$ matrices with a unique location of a nonzero entry and the diagonal is always 0. These added constraints ensure that the solution of X in our SDP remains a diagonal solution

Note that r refers to columns of A^T , which are rows of A